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Research Topic: Investigating Acoustic Detection of Lymphatic Fluid Movement for frequent Monitoring of Lymphedema

Title: Investigating Acoustic Detection of Lymphatic Fluid Movement for frequent Monitoring of Lymphedema

1. Introduction (literature review)

Lymphedema is swelling, caused by a damaged or blocked lymphatic system. It occurs in the lymphatic system, a network of vessels located throughout the human body, specifically in tissues [1].

Core problem: One of the most frequent causes of lymphedema is the after effect of cancer treatment. There are many instances where lymph vessels are damaged by cancer treatment, and if not checked frequently, impaired lymphatic drainage can progress and lead to the accumulation of lymphatic fluid in the surrounding tissues. Therefore, there is a need for accessible methods that can frequently monitor lymphatic fluid movement and identify changes in lymphatic function before the condition becomes more severe [1].

Research gap: The cost and specialized nature of existing methods limit their use for simple, repeated, and (financially) accessible monitoring. This creates an opportunity to investigate whether the underlying acoustic signal produced by ultrasound could instead be analyzed to obtain quantitative information about fluid movement without requiring a full diagnostic image so that the method may be simpler and therefore less-cost and non-radiation, since frequent radiation on top of the radiation therapy from the treatment has the potential to damage living tissue and alter cellular DNA [2].

Moreover, the method could easily be much cheaper considering a medical monitor is no longer needed; a medical monitor can cost a minimum of $2,000 dollars, equaling the high-end price of a standard consumer office monitor [18]. Ultrasound imaging relies on variations in grayscale to visualize structures such as fluid, tumors, and muscle abnormalities, requiring medical imaging systems to be calibrated according to standards such as DICOM Part 14 [17]. In contrast, a system that focuses only on detecting and analyzing returning acoustic signals would not need to generate a full diagnostic image, potentially reducing hardware and processing requirements and enabling a lower-cost approach to monitoring lymphatic fluid movement.

2. Research aims

1. Explain how lymphatic fluid is transported and how impaired drainage causes lymphedema
2. Investigate whether or not acoustic signals can detect the rate of fluid movement in the lymphatic vessels.
3. Propose a low-cost, minimal medical consequence experimental and technical treatment for lymphedema

3. Research question

Can the rate of lymphatic fluid movement be detected using a low-cost, non-invasive acoustic method without using a radioactive tracer or producing a full diagnostic image?

4. Literature Review

4.1 Lymphedema

The lymphatic system helps collect and transport excess fluids from tissues; the excess tissue fluid turns into excess fluid collected by the lymphatic vessels is called lymph [1]. The lymphatic system then removes the lymph so that the excess fluid does not accumulate; if the system was impaired or blocked, it cannot transport the lymph effectively. When lymphatic dysfunction causes persistent accumulation of lymph and associated tissue changes, lymphedema develops.

Cancer surgery can disrupt lymph vessels in the area that is being treated since lymph vessels are present throughout the body; and in some cases, lymph nodes even have to be removed, meaning there are less functioning lymph vessels which can reduce pathways for lymph to drain and thus a higher chance of accumulating in existing lymph vessels. Secondly, there is also radiation therapy, which can damage lymphatic vessels and cause scarring/fibrosis around lymphatic tissues, which can interfere with lymph transport. Often, the treatments are combined, worsening the effect of the treatment on the lymph vessels. Cancer treatment can cost a lot of money; thus, there should be a more financially and safely accessible method for frequent check-ups.

4.2 Existing methods

Several existing methods can be used to assess lymphedema and lymphatic dysfunction. Bioimpedance spectroscopy (BIS) passes a small electrical current through the body to estimate extracellular fluid and can detect changes in tissue fluid that may occur before obvious swelling [4]. Ultrasound can examine soft tissues and lymphatic structures, including tissue characteristics and superficial lymphatic vessels, while lymphoscintigraphy uses a small radioactive tracer and specialized imaging equipment to track lymphatic drainage and assess lymphatic function [6] [7]. These methods therefore provide different types of information, including changes in tissue fluid, structural characteristics, and lymphatic drainage. However, they generally require specialized equipment and, in the case of lymphoscintigraphy, radioactive tracers [5].

Despite the information provided by these methods, there remains a need for a method that can monitor fluid movement in a simpler and more accessible way. Bioimpedance spectroscopy provides information about changes in tissue fluid but does not directly measure the rate of fluid movement, while conventional ultrasound generally focuses on producing and interpreting images of tissue or lymphatic structures [8] [9]. Lymphoscintigraphy can assess lymphatic drainage, but its use of radioactive tracers and specialized imaging equipment makes it less suitable for frequent monitoring. Ultrasounds are able to detect lymphoedema, but its imaging capability makes it cost more than it would without it because it would require heavier programming and manufacturing investments, making the machinery itself inherently expensive [3].

4.3 Ultrasounds

Ultrasound is particularly relevant to the investigation because it can provide information about structures and movement within tissue. Conventional ultrasound transmits high-frequency sound waves into the body and receives echoes that are reflected or scattered by structures within the tissue [10]. These returning echoes can then be processed to produce an image, allowing structures such as vessels to be visualized [10]. However, the information contained within the returning acoustic signal is not limited to image formation because before turning into an image, it travels as acoustic waves [10]. Doppler ultrasound, for example, uses changes in echoes produced by moving scatterers to obtain information about motion and fluid velocity [10]. This

demonstrates that movement within a fluid can be detected through changes in the acoustic signal itself, rather than relying solely on a visual representation of the structures being examined. Therefore, ultrasound provides a potential basis for investigating fluid movement through analysis of its returning acoustic signals.

4.4 Acoustic scattering relation to fluid movement

The detection of fluid movement through an acoustic signal depends on the interaction between ultrasound waves and materials within the fluid. When acoustic scatterers are suspended in a fluid, ultrasound interacts with these particles and produces returning scattered signals; as the scatterers move through the ultrasound beam, their movement can cause measurable changes in the returning signal [11]. This principle is relevant to ultrasound flow phantoms, which provide controlled experimental models for investigating the relationship between fluid movement and acoustic measurements. Previous ultrasound research has demonstrated the use of moving scatterers to obtain information about fluid velocity, while flow phantoms can be used to validate and characterize these measurements under controlled conditions [12]. Therefore, a flow phantom containing controlled acoustic scatterers can provide a suitable model for investigating whether changes in fluid flow produce measurable changes in an acoustic signal.

5. Proposal

A new, simpler technological method should be established in technology to minimize financial and medical consequences. Instead of using the returning ultrasound echo primarily to construct an image, the proposed system would analyze the echo as a signal to extract information about fluid movement. In a fluid containing acoustic scatterers, the ultrasound interacts with these scatterers, and their movement changes the returning signal. Existing ultrasound research has demonstrated that moving scatterers can be used to monitor fluid velocity using Doppler techniques, while ultrasound flow phantoms are routinely used to validate and characterize flow measurements [17]. Therefore, the proposed experiment does not need to establish from scratch that ultrasound can detect motion; instead, it investigates whether this established physical principle can be simplified into a lower-complexity, signal-based measurement of fluid flow. The experiment will determine whether measurable characteristics of the returned echo can be correlated with independently calibrated flow rates through a lymphatic-vessel analogue, providing information needed to develop the proposed monitoring technology. Because the proposed system does not need to generate a full diagnostic ultrasound image [17], it would require less complex hardware and signal processing than a conventional clinical ultrasound system—therefore, a more cost-effective machine hence a more cost-effective treatment.

6. Experiment

6.1 Relevance to proposal/expected outcome: This experiment will investigate whether the existing capabilities of ultrasound can be simplified into a low-cost method for detecting fluid movement, providing a basis for determining whether such a sensing principle could eventually be developed into a wearable system for monitoring lymphatic function. The study is expected to determine whether characteristics of the reflected ultrasound signal can be used to identify and quantify changes in fluid flow through the vessel analogue. By comparing acoustic measurements with independently calibrated flow rates, the experiment will provide information on which signal characteristics, such as amplitude, are most consistently associated with fluid

movement. These findings will inform the design of the proposed low-cost acoustic monitoring system by identifying the most useful signal features for detecting and estimating changes in fluid transport. If a consistent relationship is established, the results could provide a proof of concept for replacing conventional image-based ultrasound analysis with simpler signal-based measurements. The findings will also identify technical limitations and parameters that would need to be considered when developing the system into a future device for monitoring lymphatic transport.

6.2 Model: The experiment will simulate a small lymphatic vessel within a soft-tissue analogue made from agar. A small flexible tube will represent the lymphatic vessel, with water serving as a model of fluid for lymph. A single-element circular ultrasound transducer will be positioned consistently against the phantom to transmit and receive acoustic waves. The **independent variable** will be the rate of fluid movement through the vessel's analogue, with 5 IV values: 0, 2, 4, 6, 8mL/min; the IV values are controlled by adjusting a controllable fluid pump. The model fluid will contain acoustic scatterers suspended within it, representing the particles naturally suspended in lymph that are carried by lymphatic fluid movement. The acoustic particles will be represented by microcrystalline cellulose because their physical acoustic properties differ from those of the surrounding fluid, allowing it to scatter ultrasound [13]. As the flow rate changes, these scatterers move through the ultrasound beam, producing changes in the returning acoustic signal that can be analyzed. The **dependent variable** will be the characteristics of the acoustic signal detected by the transducer, such as changes in signal amplitude or intensity over time. The depth and position of the vessel, transducer position, phantom material, fluid type, temperature, and measurement duration will be controlled to ensure that the experiment is strictly focused on investigating the relationship between the independent and dependent variable.

6.2.1 Signal Measurement: The fluid movement will be measured by recording changes in the acoustic signal amplitude over time. First, a piezoelectric circular transducer will transmit a short ultrasound pulse into the gel phantom. The transducer will receive returning echoes produced by reflections and scattering within the phantom, including scattering from the suspended particles in the flowing model fluid. These returning waves cause the piezoelectric element to generate a small electrical voltage, which is amplified by an ultrasound receiver. The amplified electrical signal will then be sent to a data-acquisition device (DAQ) with an imbedded ADC component (Analog-to-Digital Convertor) for digitization and recording, allowing the signal amplitude to be analyzed over time.

6.3 Calibration System: The relationship between flow rate and echo amplitude will then be examined using a scatter plot and Pearson's correlation coefficient 'r'. A value of 'r' close to +1 or −1 indicates a strong linear relationship, while a value close to 0 indicates a weak or negligible linear relationship [14]. Regression analysis will be used to determine how well changes in the acoustic measurement can explain variation in flow rate. This will determine whether the acoustic signal has the potential to distinguish and estimate differences in fluid flow rather than simply detect whether movement is present.

The mean and standard deviation will be calculated across repeated trials for each flow condition to assess measurement consistency. A smaller standard deviation would indicate that the measurements are more closely clustered around the mean, while a larger standard deviation

would indicate greater variability between trials. This would help identify potential outliers and asses the consistency of measurements [16].

6.3.1 Independent Variable: Before acoustic measurements are analyzed, the actual flow rate of the model fluid will be independently determined to provide a reference value. A known volume of fluid will be collected over a measured period of time, allowing the volumetric flow rate to be calculated using $Q = \frac{V}{t}$ (flow=velocity over time) [15].

6.3x .2 Dependent Variable: The ultrasound transducer will record the returning acoustic signal produced by reflections and scattering within the phantom. The recorded signal will be represented as a waveform showing changes in signal amplitude over time. For each trial, the amplitude of the returning acoustic signal will be extracted from the recorded waveform. Repeated measurements will be taken at each flow condition, and the resulting amplitude measurements can be averaged to obtain a representative acoustic amplitude for that condition. This acoustic amplitude will then be used as the dependent variable when comparing the acoustic response with the calibrated flow rate.

6.6 Limitation: A major limitation of the proposed experiment is that the model of fluid does not fully replicate the physical and biological properties of actual lymph. Although the selected liquid can be used to simulate fluid movement and provide a controllable medium for testing the acoustic signal, differences in composition and acoustic properties may affect how ultrasound interacts with the fluid compared with real lymph. Additionally, the agar phantom cannot fully reproduce the complex structure of the human body, which contains multiple layers of skin, fat, muscle, connective tissue, and other structures that can affect ultrasound transmission and reflection. Therefore, the results obtained from the phantom may not directly represent how the proposed acoustic sensing method would perform in the human body, and further testing with increasingly realistic tissue models would be required before considering practical or clinical applications.

7. Conclusion

This project investigates the feasibility of using acoustic signals to detect and quantify fluid movement through a lymphatic-vessel analogue as a potential foundation for more accessible monitoring of lymphedema. By using an agar-based tissue phantom, a flexible vessel analogue, controlled fluid flow, and a single-element ultrasound transducer, the experiment will examine whether measurable characteristics of the returning acoustic signal vary with known volumetric flow rates. Independent flow measurements will be used to calibrate and evaluate the relationship between flow rate and the selected acoustic signal characteristics.

The proposed experiment serves as a proof-of-concept investigation into whether the underlying acoustic signal used in ultrasound can provide quantitative information about fluid movement without requiring full image formation, thus verifying the ability of the bigger idea: the proposed technology. If a consistent relationship is identified, the findings could support further development of a simpler and potentially lower-cost acoustic monitoring system. Further research using more physiologically realistic lymphatic models and more complex flow conditions would be required before considering clinical applications.

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